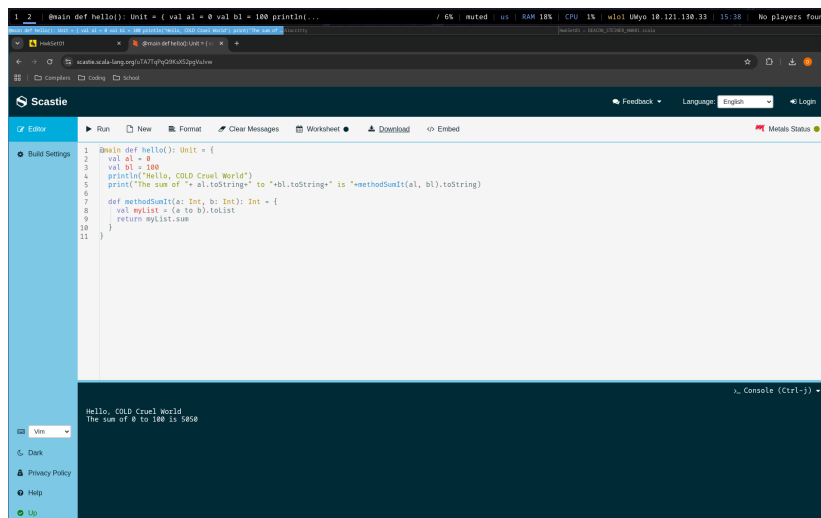


1 SCASTIE

Here I used SCASTIE to run the provided program.



The screenshot shows the SCASTIE web application. At the top, there's a status bar with system information like '0% muted', 'RAM 18%', 'CPU 1%', and a timestamp '10:12:130.23'. Below this is a browser address bar showing the URL 'scastie.sc-lang.org/u/TATq928x0k0pyshuww'. The main interface has a dark theme. On the left, there's a sidebar with 'Run', 'New', 'Format', 'Clear Messages', 'Worksheet', 'Download', and 'Embed' buttons. Below these are 'Buddi Settings', 'Vim', 'Dark', 'Privacy Policy', 'Help', and 'Up' links. The central area contains Scala code:

```
1 @main def hello(): Unit = {
2   val a1 = 0
3   val b1 = 100
4   println("Hello, COLD Cruel World")
5   print("The sum of " + a1.toString + " to " + b1.toString + " is " + methodSum1(a1, b1).toString)
6 }
7
8 def methodSum1(a: Int, b: Int): Int = {
9   val myList = (a to b).toList
10  return myList.sum
11 }
```

 At the bottom right, a 'Console (Ctrl-J)' panel shows the output:

```
Hello, COLD Cruel World
The sum of 0 to 100 is 5050
```

Figure 1: SCASTIE program output for initial program

Getting SCASTIE running was no real issue for me. It was easy to find and made writing code very easy.

The screenshot shows the Scastie web interface. The top bar displays system information: 6% VDL, 115% us, RAM 24%, CPU 1%, w101 18.121.138.33, 09:58, No players found. The browser tabs include 'Scastie', 'Scala Playground', 'Scala Docs', and 'Scala Playground'. The main editor contains the following Scala code:

```
1 @main def hello(): Unit = {
2   val a1 = 21
3   val b1 = 137
4   println("Hello, COLD Cruel World")
5   print("The sum of " + a1.toString + " to " + b1.toString + " is " + methodSum(a1, b1).toString)
6 }
7 def methodSum(a: Int, b: Int): Int = {
8   val myList = (a to b).toList
9   return myList.sum
10 }
11 }
```

The console output at the bottom shows:

```
Hello, COLD Cruel World
The sum of 21 to 137 is 9243
```

Figure 2: SCASTIE program output for updated sum

This program calculates the sum of integers from 21 through 137. The correctness of this program can be verified by looking at the closed-form formula for the sum of consecutive integers.

$$\sum_{n=21}^{137} n = \frac{137(137 + 1)}{2} - \frac{20(20 + 1)}{2}$$

which comes out to:

$$= \frac{18906}{2} - \frac{420}{2} = 9453 - 210 = 9243$$

```
1 2 3 HaSet01 - DEACON_STEINER_HM01.scala / 6% | VOL 115% | us | RAM 27% | CPU 2% | w101 10yo 18.121.138.33 | 10:14 | No players found  
DEACON_STEINER_HM01.scala  
def methodSum(a: Int, b: Int): Int =  
  ((b * (b+1)) / 2) - (((a-1) * a) / 2)  
  
@main def hello(): Unit = {  
  val a1 = 21  
  val b1 = 237  
  println("The sum of all integers from " + a1.toString + " to " + b1.toString + " is " + methodSum(a1, b1).toString)  
}
```

Figure 3: Scala program on local machine

```
1 2 3 Alacritty / 6% | VOL 115% | us | RAM 28% | CPU 1% | w101 10yo 18.121.138.33 | 10:17 | No players found  
coursier-2015/HaSet01 on main [17] via v3.0.4  
> scala DEACON_STEINER_HM01.scala  
The sum of all integers from 21 to 237 is 9243  
coursier-2015/HaSet02 on main [17] via v3.0.4  
>
```

Figure 4: Output for local program

Installing Scala wasn't difficult, although I did run into a small issue after getting it installed initially. Scala and Coursier had to be manually added to my PATH so that their commands would be recognized by the terminal.